QUOTATION

Kitchen Airflow Measurement, Analysis & Balancing Report

Quotation No:	MES/QTN/AZD/2026-0315	**Date:**	15 March 2026
Reference:	C4087-P16568(Q2)-09-24	**Valid Till:**	14 April 2026

1. CLIENT DETAILS

Client:	Azadea Group
Attention:	Mr. Mohamed Siyath Mohamed Ahas - Store Services Administrator
Project:	Butcher Shop & Grill Restaurant (LG010B) at Place Vendome Mall
Subject:	Kitchen Negative Pressure Issue - Airflow Measurement, Analysis & TAB Report

2. BACKGROUND

The kitchen at Butcher Shop & Grill Restaurant is experiencing excessive cold air intrusion from the dining area, adversely affecting food plating temperatures and kitchen comfort conditions. The HVAC design consultant (Al-Kashaf International) has confirmed that the design maintains approximately 10–13% kitchen negative pressure in accordance with applicable standards, and has recommended engaging a certified TAB company to verify and adjust the airflow rates to match the approved design and hood calculations.

As per the hood calculation report (Ref: Al-Kashaf International, dated 05-Nov-2024), the kitchen is served by five (5) exhaust hoods with a total design exhaust of 3,759 L/s, make-up air provisions through double-skin hoods (Hoods 01 & 02), and a Fresh Air Handling Unit (FAHU) delivering 2,280 L/s to the kitchen. The design net kitchen negative is -479 L/s.

3. SCOPE OF WORK

Metri Engineering Services shall carry out the following scope of work:

Phase 1: Field Airflow Measurements

- Measure exhaust airflow rate (CFM/L/s) at each of the five (5) kitchen hoods using calibrated instruments (balometer/anemometer):
  - Hood No. 01 – Custom Hood, Double Skin (Design: 460 L/s exhaust, 415 L/s make-up)
  - Hood No. 02 – Custom Hood, Double Skin (Design: 905 L/s exhaust, 585 L/s make-up)
  - Hood No. 03 – Custom Hood, Single Skin (Design: 1,020 L/s exhaust)
  - Hood No. 04 – Custom Hood, Single Skin (Design: 1,020 L/s exhaust)
  - Hood No. 05 – Custom Hood, Single Skin (Design: 354 L/s exhaust)
- Measure make-up air supply rate at Hoods 01 and 02 (double-skin integrated supply)
- Measure FAHU supply airflow to the kitchen (Design: 2,280 L/s)
- Measure fresh air supply to the seating/dining area from LOD (Design: 490 L/s) and FAHU (Design: 120 L/s)
- Record differential pressure readings between kitchen and dining area using a manometer

Phase 2: Comparative Analysis

- Tabulate measured airflow values against the approved design values from the Al-Kashaf hood calculation report
- Calculate actual kitchen air balance (supply vs. exhaust) and determine the actual negative pressure ratio
- Identify deviations: over-exhausting hoods, under-performing supply, FAHU deficiencies, etc.

Phase 3: Technical Report & Recommendations

- Issue a comprehensive TAB report with all measured data, design comparisons, and deviation analysis
- Provide technical recommendations to resolve the kitchen negative pressure issue, which may include: damper adjustments, fan speed modifications, equipment replacement recommendations (if applicable), or balancing procedures
- Provide a proposed corrective action plan with prioritized steps

4. DESIGN REFERENCE DATA

As per Al-Kashaf International Hood Calculation Report (05-Nov-2024):

Description	Exhaust (L/s)	Supply (L/s)	Balance (L/s)	Skin Type
Hood No. 01	460	415	-45	Double
Hood No. 02	905	585	-320	Double
Hood No. 03	1,020	0	-1,020	Single
Hood No. 04	1,020	0	-1,020	Single
Hood No. 05	354	0	-354	Single
FAHU to Kitchen	0	2,280	+2,280	—
LOD to Dining	0	490	+490	—
FAHU to Dining	0	120	+120	—
TOTAL	**3,759**	**3,890**	**+131**	—

Design Kitchen Pressure: -479 L/s | Design Restaurant Pressure: +131 L/s

5. PRICING

S.No	Description	Amount (QAR)
1	**Kitchen Airflow Measurement, Comparative Analysis & Technical Report** Including: Field measurement of all 5 hoods (exhaust & make-up), FAHU supply verification, dining area supply verification, design vs. measured comparison, deviation analysis, and technical recommendations report with corrective action plan.	**5,000.00**
	TOTAL (Lump Sum)	**QAR 5,000.00**

(Qatari Riyals Five Thousand Only)

6. PAYMENT TERMS

- 50% advance payment upon issuance of Purchase Order (QAR 2,500.00)
- 50% upon submission and of the final TAB report (QAR 2,500.00)

7. TIMELINE

5 days

8. EXCLUSIONS & CLARIFICATIONS

1. This quotation covers airflow measurement, analysis, and reporting only. (equipment or parts replacement) are excluded and shall be quoted separately if required.
2. Access to all kitchen hoods, FAHU, and associated ductwork/grilles shall be provided by the client during the site visit.
3. Measurements shall be conducted during normal HVAC operating conditions with all hoods and equipment running.
4. Site visit scheduling is subject to restaurant operational hours and coordination with the client.
5. This quotation is valid for thirty (30) calendar days from the date of issue.

9. ACCEPTANCE

We trust the above meets your requirements. Please confirm your acceptance by issuing a Purchase Order referencing this quotation number.

For Metri Engineering Services W.L.L.	Client Acceptance (Azadea Group)
متري للخدمات الهندسية ذ.م.م ص.ب : ١٩٩٦٢ الدوحة - قطر تليفكس: ٤٤١٢٧٦٧ P.O. Box 19962, Doha-Qatar TelFax: 4412767 METRI ENGINEERING SERVICES W.L.L	Authorized Signatory Name: _____ Date: _____